Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/26/2025
PI Name:	Christov, Ivan Ph.D.		
Project Title:	Validation of a CFD Model for Gas-Liquid Flows in Packed Bed Reactors to Enable Thermo-Fluid Analysis in Microgravity		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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PI Email:	christov@purdue.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	765-496-3733
Organization Name:	Purdue University		
PI Address 1:	School of Mechanical Engineering		
PI Address 2:	585 Purdue Mall		
PI Web Page:			
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2088	**Congressional District:**	4
Comments:			
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Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
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Task Description:	The proposed use of NASA's Physical Sciences Informatics (PSI), specifically the packed bed reactor experiment (PBRE) data, will enable new research on fluid physics in microgravity conditions. Two-phase gas-liquid flows are ubiquitous in life support and thermal control systems for spacecraft, space stations, and proposed habitats on the moon and Mars. Two-phase flows are impacted by microgravity conditions because while on Earth capillary forces are easily overcome by gravitational forces, the opposite is true in low-gravity environments. Predictive modeling of these flows is challenging -- even for terrestrial applications -- and the lack of predictive models limits the ability to scale up systems to sizes required for NASA missions. In the first year of the project, we will develop a computational fluid dynamics (CFD) approach for predicting dispersed gas-liquid flows. The CFD approach will be validated against the PBRE data from the PSI, specifically visual images of gas-liquid flows from videos, as well as corresponding pressure drop information across disparate flow regimes. First, a suitable 3D flow geometry will be constructed from the specifications of the PBRE. Then, interface-resolved simulations will be performed of gas-liquid flow through a packed bed. Using supercomputing resources at Purdue, we will perform large-scale simulations of gas-liquid flows across regimes (bubbly, slug, core-annular, stratified, etc.) at full and reduced gravity conditions. The gas-liquid interface dynamics, as we as the flow-wise pressure gradient, will be computed for different Reynolds numbers and gas volume fractions and compared to the PBRE datasets in the PSI. Tracking the complex growing, merging, and rupturing gas-liquid interfaces in simulations is challenging. To address scale-up, two-fluid models, in which the gas and liquid phases are considered to be interpenetrating continua, have been proposed. Two-fluids models reduce computational cost by removing the need to track interfaces; however, they require calibration of parameters to become predictive. Specifically, a correlation for the interphase drag force must be developed. Previous work has shown that the dimensionless parameter space consists of the Suratman number (a modified gas Reynolds number with the velocity set by the ratio of surface tension to viscosity) and the ratio of the gas and liquid phases' Reynolds numbers. In second year of the project, we will calibrate two-fluid models, under the steady 1D flow assumption, by fitting the interphase drag coefficient to data from the PSI. In turn, two-fluid models will enable large-scale simulations, with unresolved pore-scale dynamics, towards scale up for packed-bed reactor properties. Specifically, two-fluid model simulations using the novel correlations learned from the PBRE datasets in the PSI will help determine the boundaries for flow regime transitions in the parameter space, leading to new regime diagrams, and specifically sharper transition boundaries between regimes, that address gaps in the current understanding. While the flow physics alone is challenging to predict, such two-phase gas-liquid flows are often coupled with heat transfer and chemical reactions. In parallel, we propose to evaluate the feasibility of ground-based thermo-fluid experiments in a packed bed reactor against which to validate the extended models. In future work, correlations for heat transfer coefficients for gas-liquid flows in packed beds can be calibrated against novel ground-based experiments, which we will explore based on the design of the PBRE. Ultimately, the proposed research will provide tools for accurately modeling coupled thermal-fluid systems for both terrestrial and low-gravity applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed use of NASA's Physical Sciences Informatics (PSI) data repository, specifically the packed bed reactor experiment (PBRE) data, will enable new research on fluid physics across different gravity conditions. The 2011 National Academies decadal survey lists heat and mass transfer in porous media as a recommended research direction: ``TSES6—NASA should conduct research for the development and demonstration of closed-loop life support systems and supporting technologies. Fundamental research includes heat and mass transfer in porous media under full, partial and microgravity conditions and understanding the effect of variable gravity on multiphase flow systems." The proposed research aims to fill a knowledge gap in this area, which remains ``highest priority" in the midterm assessment of the decadal survey. Two-phase gas-liquid flows are ubiquitous in life support and thermal control systems for spacecraft, space stations, and proposed habitats on the Moon and Mars, as well as terrestrial applications involving distillation, purification and separation, and catalytic reactions to enhance heat and mass transfer in convective flows, among other examples. However, gas-liquid flows through packed bed reactors have not been fully understood. Predictive modeling of these flows is challenging---even for terrestrial applications---and the lack of predictive simulation tools prohibits effective scaling up of systems to sizes required for future missions. To this end, we propose a computational fluid dynamics (CFD) approach for simulating gas-liquid flows in porous media across regimes: from bubbly, slug, core-annular flow to fully-dispersed gas phase. The CFD approach will be validated against the PBRE data from the PSI and extended to flows with heat transfer.
Task Progress:	This is the final report for the project, summarizing the total task progress. Our major accomplishments during the award were the publication of 2 peer-reviewed articles in top journals. [See the Cumulative Bibliography.] Understanding transport phenomena through porous media is essential for applications ranging from water treatment systems to heat pipes. In many of these systems, packed bed reactors (PBRs) are crucial components, and understanding and quantifying the pressure drop due to flow through the PBR is critical to effective operation. Prior works on the hydrodynamics of gas-liquid flow through a PBR under microgravity conditions have focused on characterizing the flow regime (into a bubble or pulse flow) and the corresponding pressure drops, without providing insight into the dynamics of the bubbles. Moreover, in the recent analysis of microgravity experiments on the International Space Station (ISS), Motil et al. noted that "one fundamental problem identified in the packed bed reactor experiment (PBRE) is that a minimum liquid superficial flux is needed to dislodge trapped bubbles in a porous medium". [Ed. Note: See REFERENCES below.] To better understand this fundamental problem and thus fill a knowledge gap in the literature, in our first peer-reviewed publication, we explored the bubble flow regime in-depth by investigating the influence of inertia, capillary, and buoyancy forces on a bubble's entrapment and displacement as it traverses the tortuous pore spaces of a PBR. We developed a workflow for generating the computational geometry of a PBR representative element volume (REV), and we performed simulations within this REV. To rationalize the dynamics of bubbles in PBRs at microgravity, we introduced two new dimensionless numbers that represent dynamic balances between capillary, inertial, and gravitational forces, as quantified through our computational fluid dynamics (CFD) simulations. Motivated by the lack of fundamental studies on two-phase gas-liquid flow through a PBR under microgravity

	conditions, in our second peer-reviewed publication, we developed a data-driven methodology for calibrating a gas-liquid drag closure that can be used in CFD simulations. Recent experiments conducted by NASA measured the pressure drop due to gas-liquid flow through a PBR under microgravity conditions. Based on these experiments, we developed correlations for the interphase drag in a two-fluid model (TFM). Then, under the assumption of a 1D flow, we rewrote the TFM equations with the gas-liquid drag as the only unknown. We employed data-driven calculations to determine the drag, which we correlated (via composite fits) as a function of the liquid and gas Reynolds numbers based on the superficial velocities and the Suratman number. To demonstrate the use of these correlations, we performed 2D transient, multiphase CFD laminar flow simulations in ANSYS® Fluent employing an Euler-Euler formulation. These CFD simulations predicted a pressure gradient 25% of the experimental values reported in the PSI, highlighting the validity of the proposed approach based on the Euler-Euler simulation using the proposed drag law. REFERENCES Motil BJ, Ramé E, Salgi P, Taghavi M, Balakotaiah V. "Gas-liquid flows through porous media in microgravity: The International Space Station Packed Bed Reactor Experiment." AIChE J. 2021 Jan;67(1):e17031. https://
Bibliography Type:	Description: (Last Updated: 04/28/2025)
Articles in Peer-reviewed Journals	Nagrani PP, Marconnet AM, Christov IC. "Correlations for the interphase drag in the two-fluid model of gas–liquid flows through packed-bed reactors." Industrial & Engineering Chemistry Research. 2025 Feb 24;64:5049-56. https://doi.org/10.1021/acs.iecr.4c04147 , Feb-2025